# Ismail Ifakir

PhD Student

## Profile

I am a 25-year-old PhD student passionate about data science. I'm eager to apply my expertise in data science, machine learning, and AI to impactful projects. Driven by curiosity and a commitment to professional growth, I am motivated to contribute to innovative solutions and advance my skills in cutting-edge technologies.

## Education

**2024 – Present** — **Faculty of Sciences Dhar El Mahraz, Fez, Morocco**

Ph.D student (Multimodal Aspect-Based Sentiment Analysis)

**2024** — **Polydisciplinary Faculty of Nador, Morocco**

Master's in Data Sciences and Intelligent Systems

**2022** — **Polydisciplinary Faculty of Ouarzazate,Morocco**

Bachelor's in Mathematics and Computer Science

## Projects

2026 ↑ 2019

### Dynamic Website for a Private School

- Bachelor's Degree Completion Project: Realization of a Dynamic Website for a Private School. The project involved the use of technologies such as PHP, HTML, CSS, and JavaScript

### Speaker recognition

- Deep Learning: CNN

### Intrusion detection systems

- Classification of Normal and Attack messages using ML.

### Image Classification

- Diabetes detection Using CNN, VGG16, ResNet, and ViT.

### Research Laboratory Management Application

- Spring Boot, MySQL, HTML, CSS, and JavaScript

### Sentiment Analysis of Moroccan Arabic Dialect

- Master's Degree Completion Project: DrijaBERT for embedding; CNN, LSTM, BiLSTM, and GRU as base classifiers; Logistic Regression as the meta-classifier; and Streamlit for the web application.

### Generate Question from CVs

- Internship at 3D Smart Factory: Extracted entities from resumes using Named Entity Recognition (NER) with spaCy and RoBERTa.

## Contact


**Email**
ismail.ifakir@usmba.ac.ma

**Phone**
+212 681 530 755

**Address**
Fez-Meknes

**ORCID**
Ismail IFAKIR

**LinkedIn**
Ismail-IFAKIR

**Website**
Ismail-IFAKIR

**Github**
Ismail-IFAKIR


## Teaching Exp.

### Teaching Assistant in Artificial Intelligence — Higher School of Technology of Fez (HSTF)

- Taught 60 hours of the module *Artificial Intelligence and Industrial Processes* for the *Process Engineering* program during the 2025/2026 academic year.

### Teaching Assistant in Artificial Intelligence — Faculty of Sciences and Techniques of Al Hoceima (FSTH)

- Delivered 10 hours of instruction in the module *Artificial Intelligence* for the *Data Engineering and Software Development (IDDL, S6)* program during the 2025/2026 academic year.

## Skills

- Python,Java,C,JavaScript,HTML/CSS
- Node.js, Flask, Streamlit
- SQL/NOSQL
- ML/DL/NLP/LLMs/Multimodal/RAG
- Generative AI/ Agentic AI
- MongoDB, Cassandra, ETL
- NumPy, Pands, Matplotlib, NLTK
- Scikit-learn,TensorFlow,Keras,PyTorch

- Developed a question generation pipeline based on the extracted entities using Google Gemini.

### Multimodal Aspect-Based Sentiment Analysis

- RoBERTa for text embedding, ViT for image embedding, and LSTM as the classifier.

### Chatbot using RAG

- Developed a complete RAG (Retrieval-Augmented Generation) pipeline from scratch using LangChain. This included ingesting raw data, preprocessing, creating embeddings, and storing them in a vector database, providing the foundation for powerful RAG applications.
    1. Building a retrieval query pipeline with GROQ.
    2. Integrating a vector database into the pipeline.

### Multimodal RAG

- Built a Multimodal RAG pipeline with LangChain using PDF data sources (text + images), leveraging OpenAI's multimodal LLM `gpt-4.1` and Ollama's multimodal model `gemma3:4b`.

### Agentic AI for Multiple Data Sources

- Developed advanced RAG-based Agentic AI using `Ollama LLaMA2`, integrating multiple data sources (e.g., Arxiv, Wikipedia, and others) with agents, tools, toolkits, and an agent executor.

## Publications

### An Attention-Driven Feature Fusion Approach for Multimodal Aspect-Based Sentiment Analysis

- **Journal Paper — Q1**
- Authors: Ismail Ifakir, El Habib Nfaoui, Abderrahim Zannou, and Asmaa Mourhir
- Published in *Big Data and Cognitive Computing* (BDCC), MDPI.
- Proposed ADFF, a three-stage hierarchical attention fusion approach for multimodal aspect-based sentiment analysis.

### An Approach Based on Named Entity Recognition and Semantic Analysis for Recruitment Efficiency and Optimization

- **Journal Article — Q3**
- Authors: Ismail Ifakir, Noureddine Mohtaram, El Habib Nfaoui, Abderrahim Zannou, and Mohammed El Hassouni
- Proposed an AI-based recruitment pipeline using NER, transformer-based semantic matching, and LLM-generated interview questions.

### Attention-Guided Fusion for Multimodal Aspect-Based Sentiment Analysis

- **Conference Paper**
- Authors: Ismail Ifakir, El Habib Nfaoui, and Abderrahim Zannou
- Proposed a two-stage attention-guided fusion framework for aspect-aware text-image sentiment analysis.

## Certifications

- `Machine Learning`
- `Deep Learning`
- `Natural Language Processing`
- `Transformer Models`
- `Generative AI`
- `JSIIA 2025 Certificate`
  Communication certificate for presenting: *Sentiment Analysis of Moroccan Arabic Dialect*. Ben M'sik Faculty of Sciences, May 2025.
- `ICOA 2025 Certificate`
  Communication certificate for presenting: *Attention-Guided Fusion for Multimodal Aspect-Based Sentiment Analysis*. Kenitra, Morocco, October 2025.

## Languages

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|---|---|
| **Tamazight** | Native |
| **Arabic** | Native |
| **English** | Advanced |
| **French** | Intermediate |